1 F^n

Recall that F stands for $\mathbb{R}$ or $\mathbb{C}$.

Definition 1.1 (F^n)**.** For each positive integer n, define F^n to be the set of all ordered n-tuples of elements of F. That is,

$$F^n := \{(x_1,\ldots,x_n) : x_1,\ldots,x_n \in F\}.$$

We often denote an element of F^n by a single letter, such as x. That is, we let

$$x = (x_1,\ldots,x_n).$$

Then $x_k \in F$ is called the *k-th component* (or kth *coordinate)* of x. Two elements of F^n are *equal* if and only if the corresponding coordinates are all equal. That is, $(x_1,\ldots,x_n) = (y_1,\ldots,y_n)$ if and only if $x_1 = y_1,\ldots,x_n = y_n$.

Example 1.2.

(a) An element of F^1 is a 1-tuple (x), with $x \in F$. We identify (x) with x, so that $F^1 = F$.

(b) $\mathbb{R}^2 = \{(x,y) : x,y \in \mathbb{R}\}$ is the plane.

(c) $\mathbb{R}^3 = \{(x,y,z) : x,y,z \in \mathbb{R}\}$ is three-dimensional space.

(d) $\mathbb{C}^4 = \{(z_1,z_2,z_3,z_4) \in \mathbb{C}\}$ is the set of all ordered 4-tuples of complex numbers.

Definition 1.3 (Addition in F^n)**.** *Addition* in F^n is defined by adding the corresponding components:

$$(x_1,\ldots,n) + (y_1,\ldots,y_n) = (x_1+y_1,\ldots,x_n+y_n).$$

Proposition 1.4. *Addition in F^n has satisfies the same properties as addition in F, namely*

Properties of Addition
(A1) *(Commutativity) $x+y = y+x$ for all $x,y \in F^n$.*
(A2) *(Associativity) $(x+y)+z = x+(y+z)$ for all $x,y,z \in F^n$*
(A3) *(Identity) There exists an element $0 \in F^n$ such that $0+x = x$ for every $x \in F^n$.*
(A4) *(Additive Inverses) To each $x \in F^n$ there exists an element $-x \in F^n$ such that $x+(-x) = 0$.*

Proof. To prove (A1), let $x = (x_1,\ldots,x_n)$ and $y = (y_1,\ldots,y_n)$. Then

$$\begin{aligned} x+y &= (x_1+y_1,\ldots,x_n+y_n) \\ &= (y_1+x_1,\ldots,y_n+x_n) \qquad \text{(since (A1) holds for each } x_k+y_k \in F) \\ &= y+x. \end{aligned}$$

The proof that the remaining properties hold is left as an exercise. □

1. Complete the proof of Proposition 1.4.

Note that the proofs of the properties which follow from (A1)-(A4), such uniqueness of the additive identity 0, the cancellation law, etc., all carry over immediately to F^n.

Let's now turn to multiplication. Suppose we try to define multiplication on F^n similar to the way we defined addition, namely by multiplying the corresponding coordinates together. That is,

$$(x_1,\ldots,x_n)(y_1,\ldots,y_n) = (x_1y_1,\ldots,x_ny_n).$$

This is clearly commutative and associative, and the multiplicative identity is $(1,\ldots,1)$. However, not all nonzero elements of F^n have a multiplicative inverse. For example, $(1,0,\ldots,0) \neq (0,\ldots,0) = 0$, but there is no element $(x_1,\ldots,x_n) \in F^n$ such that $(x_1,\ldots,x_n)(1,0,\ldots,0) = (x_1,0,\ldots,0) = (1,1,\ldots,1)$. Thus, this definition of multiplication does not make F^n into a field. For this reason, and others, the definition above turns out to not be useful. What does turn out to be useful, is to define the product of an element of F^n by an element of F.

Definition 1.5 (Product of an element of F and an element of F^n)**.** The product of $\alpha \in F$ and $x = (x_1,\ldots,x_n) \in F^n$ is defined by multiplying each component of x by α. That is,

$$\alpha(x_1,\ldots,x_n) := (\alpha x_1,\ldots,\alpha x_n)$$

2. Prove that, for all $\alpha,\beta \in F$ and all $x,y \in F^n$, the following properties are satisfied:

   (a) $(\alpha\beta)x = \alpha(\beta x)$

   (b) $(\alpha+\beta)x = \alpha x + \beta x$

(c) $\alpha(x+y) = \alpha x + \alpha y$

(d) $1x = x$

3. Find $x \in \mathbb{R}^4$ such that $(4,-3,1,7) + 2x = (5,9,-6,8)$.

4. Explain why there does not exist $\alpha \in \mathbb{C}$ such that

$$\alpha(2-3i, 5+4i, -6+7i) = (12-5i, 7+22i, -32-9i).$$